

Assignment

ugizwenayo Divine

September 2026

Introduction

Association Rule Learning is an unsupervised learning technique used to find patterns between items that frequently occur together. It is commonly used in market basket analysis to understand customers' purchasing behavior.

In this assignment, the Groceries dataset was analyzed using Python. The analysis focused on finding frequently purchased products, frequent itemsets, and association rules. The Apriori algorithm was used to generate the association rules.

The main measures used in the analysis were support, confidence, and lift.

1 Question 1: Understanding the Data

1.1 (a) Number of Transactions

The Groceries dataset contains **9,835 transactions**.

1.2 (b) Number of Different Grocery Items

There are **169 different grocery items** in the dataset.

1.3 (c) Top 10 Most Frequently Purchased Items

The top 10 most frequently purchased items are shown in Table 1.

Table 1: Top 10 most frequently purchased items

Rank	Item	Transactions
1	whole milk	2513
2	other vegetables	1903
3	rolls/buns	1809
4	soda	1715
5	yogurt	1372
6	bottled water	1087
7	root vegetables	1072
8	tropical fruit	1032
9	shopping bags	969
10	sausage	924

Whole milk was the most frequently purchased item, appearing in 2,513 transactions.

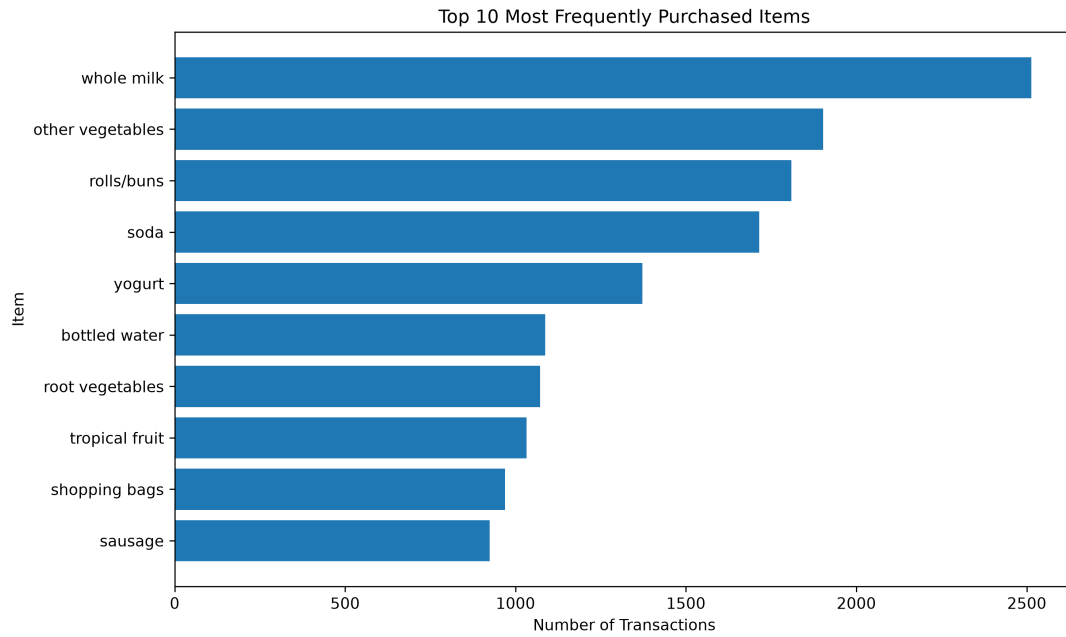


Figure 1: Top 10 most frequently purchased items

2 Question 2: Frequent Itemsets

For this question, products that appeared together in the same transaction were counted. Their support was then calculated using the total number of transactions.

2.1 (a) and (b) Frequent Itemsets and Support

The top 10 frequent two-product itemsets are shown in Table 2.

Table 2: Top 10 frequent itemsets

Rank	Itemset	Count	Support
1	other vegetables, whole milk	736	7.48%
2	rolls/buns, whole milk	557	5.66%
3	whole milk, yogurt	551	5.60%
4	root vegetables, whole milk	481	4.89%
5	other vegetables, root vegetables	466	4.74%
6	other vegetables, yogurt	427	4.34%
7	other vegetables, rolls/buns	419	4.26%
8	tropical fruit, whole milk	416	4.23%
9	soda, whole milk	394	4.01%
10	rolls/buns, soda	377	3.83%

2.2 (c) Itemset with the Highest Support

The itemset with the highest support was:

{other vegetables, whole milk}

It appeared in 736 transactions and had a support of **0.0748 (7.48%)**.

2.3 (d) Meaning of Support

Support tells us how frequently an itemset appears in all transactions. It can be calculated as:

$$Support = \frac{\text{Number of transactions containing the itemset}}{\text{Total number of transactions}}$$

For example, the support of 0.0748 for {other vegetables, whole milk} means that **7.48% of all transactions contained both products**.

A higher support means that the itemset appears more frequently in the dataset.

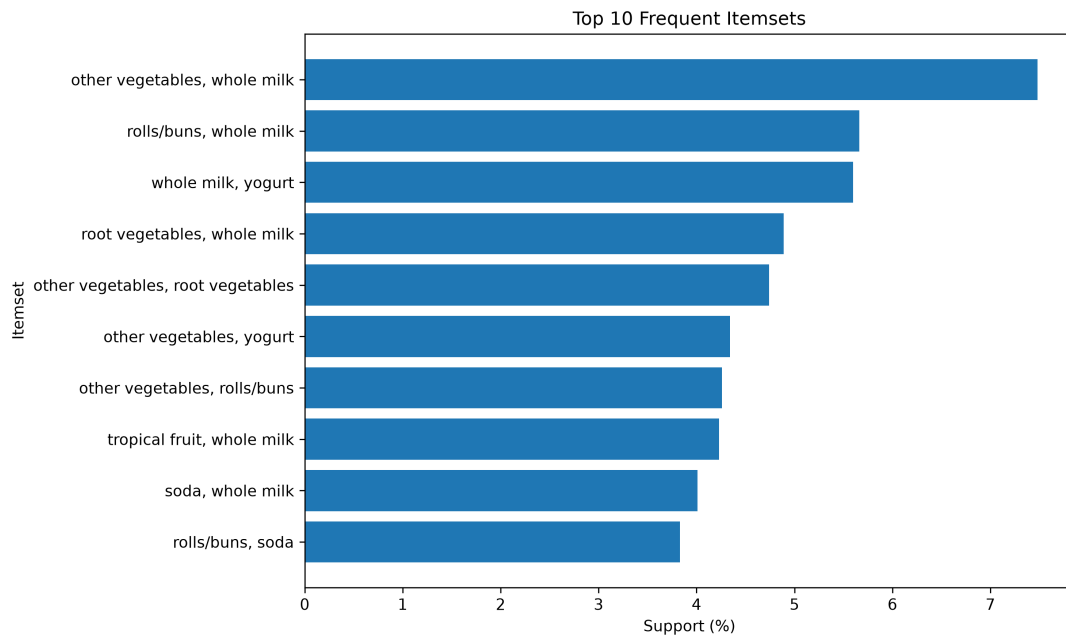


Figure 2: Top 10 frequent itemsets

3 Question 3: Association Rules

The Apriori algorithm was used to generate the association rules. A minimum support of 0.01 and a minimum confidence of 0.10 were used.

The analysis generated:

- 333 frequent itemsets
- 460 association rules

The following table shows 10 of the generated association rules.

Table 3: Association rules generated from the dataset

Antecedent	Consequent	Support	Confidence	Lift
curd	whole milk, yogurt	0.0101	0.1889	3.3723
whole milk, yogurt	curd	0.0101	0.1797	3.3723
citrus fruit, other vegetables	root vegetables	0.0104	0.3592	3.2950
other vegetables, yogurt	whipped/sour cream	0.0102	0.2342	3.2671
whipped/sour cream	other vegetables, yogurt	0.0102	0.1418	3.2671
root vegetables	other vegetables, tropical fruit	0.0123	0.1129	3.1448
other vegetables, tropical fruit	root vegetables	0.0123	0.3428	3.1448
root vegetables	beef	0.0174	0.1595	3.0404
beef	root vegetables	0.0174	0.3314	3.0404
citrus fruit, root vegetables	other vegetables	0.0104	0.5862	3.0296

The **antecedent** is the item or group of items on the left side of the rule. The **consequent** is the item or group of items on the right side.

Support shows how often the complete combination occurs. Confidence shows how often the consequent occurs when the antecedent is present. Lift shows how strong the association is compared with the normal frequency of the consequent.

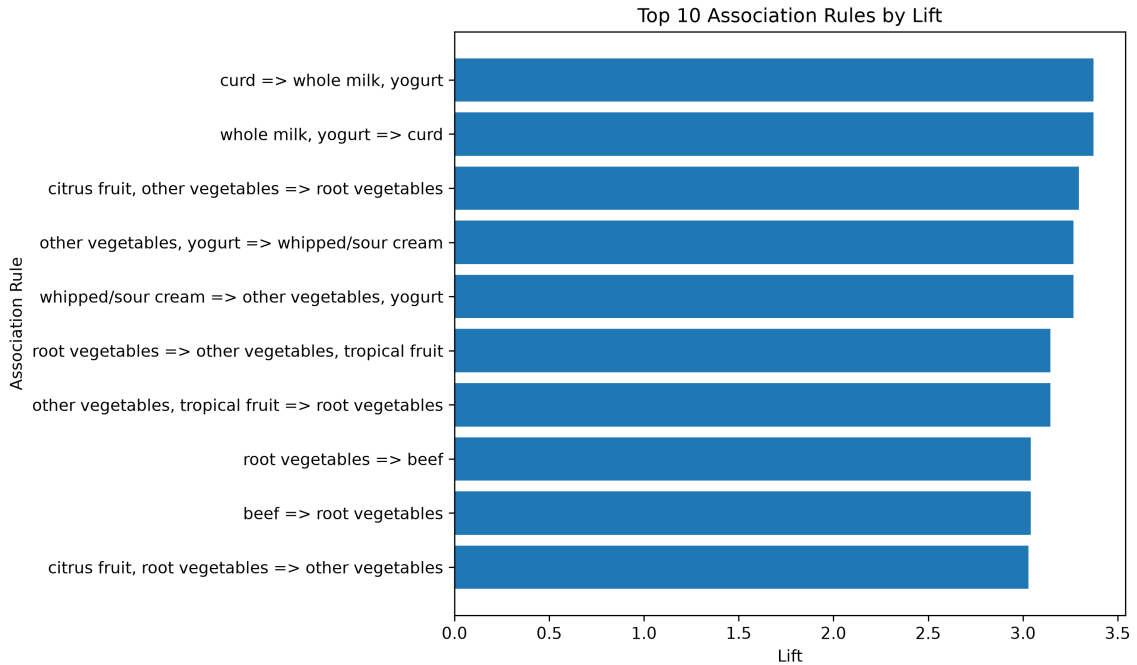


Figure 3: Top 10 association rules by lift

4 Question 4: Understanding the Measures

Three rules generated in Question 3 were selected for further explanation.

Table 4: Three selected association rules

Rule	Support	Confidence	Lift
curd $\rightarrow$ whole milk, yogurt	0.0101	0.1889	3.3723
citrus fruit, other vegetables $\rightarrow$ root vegetables	0.0104	0.3592	3.2950
citrus fruit, root vegetables $\rightarrow$ other vegetables	0.0104	0.5862	3.0296

For the rule **{curd} $\rightarrow$ {whole milk, yogurt}**, the support of 0.0101 means that 1.01% of all transactions contained all three products. The confidence of 0.1889 means that 18.89% of transactions containing curd also contained whole milk and yogurt. The lift of 3.3723 shows a positive association.

For the rule **{citrus fruit, other vegetables} $\rightarrow$ {root vegetables}**, the support is 0.0104, or 1.04%. The confidence is 0.3592, meaning that 35.92% of transactions containing citrus fruit and other vegetables also contained root vegetables. Its lift of 3.2950 also shows a positive association.

For the rule **{citrus fruit, root vegetables} $\rightarrow$ {other vegetables}**, the support is 0.0104, or 1.04%. The confidence is 0.5862, meaning that 58.62% of transactions containing citrus fruit and root vegetables also contained other vegetables. Its lift of 3.0296 shows a positive association.

5 Question 5: Finding Interesting Rules

The 460 rules generated in Question 3 were compared using support, confidence, and lift.

5.1 (a) Rule with the Highest Confidence

The rule with the highest confidence was:

$$\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$$

Its confidence was **0.5862 (58.62%)**.

5.2 (b) Rule with the Highest Lift

The rule with the highest lift was:

$$\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$$

Its lift was **3.3723**.

5.3 (c) Rule with the Highest Support

The rule with the highest support was:

$$\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$$

Its support was **0.0748 (7.48%)**.

5.4 (d) Are These the Same Rule?

No. The three measures identified different rules.

The highest-confidence rule was $\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$, while the highest-lift rule was $\{\text{curd}\} \rightarrow \{\text{whole milk, yogurt}\}$. The highest-support rule was $\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$.

This happens because support, confidence, and lift measure different properties of an association rule.

6 Question 6: Interpretation of a Rule

For this question, one of the rules generated in **Question 3** was selected:

$$\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$$

The results from Question 3 were:

- Support = **0.0104 (1.04%)**
- Confidence = **0.5862 (58.62%)**
- Lift = **3.0296**

In simple terms, the rule means that customers who purchase citrus fruit and root vegetables also tend to purchase other vegetables.

The support of 1.04% means that 1.04% of all transactions contained all three products.

The confidence of 58.62% means that among transactions containing citrus fruit and root vegetables, 58.62% also contained other vegetables.

The lift of 3.0296 is greater than 1, showing a positive association. It means that other vegetables occur in these transactions at about three times their baseline rate.

This shows a relatively strong association in the dataset. However, it does not mean that buying citrus fruit and root vegetables causes customers to buy other vegetables.

7 Question 7: Business Application

Association rules can help a supermarket understand which products customers tend to purchase together.

For example, the rule

$$\{\text{citrus fruit, root vegetables}\} \rightarrow \{\text{other vegetables}\}$$

could be used to recommend products to customers. If a customer buys citrus fruit and root vegetables, the supermarket could recommend other vegetables. The rule has a confidence of 58.62% and a lift of 3.0296.

The supermarket could also use the results for:

- **Product placement:** Related products could be placed closer together.
- **Product recommendations:** An online supermarket could recommend products based on what a customer has already selected.
- **Promotions and bundles:** Products that frequently occur together could be included in special offers.
- **Inventory planning:** The supermarket could pay attention to the stock of products that are commonly purchased together.

Overall, the rules provide useful information for recommendations, product placement, promotions, and inventory planning.

8 Question 8: Understanding Association

8.1 (a) Does a Strong Association Mean Causation?

No. A strong association does not mean that one product causes another product to be purchased.

Association rule mining only shows that products frequently occur together in the transaction data. For example, the rule used in Question 6 has a confidence of 58.62% and a lift of 3.0296. These values show an association, but they do not prove that buying citrus fruit and root vegetables causes a customer to buy other vegetables.

There may be other reasons why the products are purchased together. For example, a customer may be buying several ingredients for the same meal.

8.2 (b) What Does High Confidence but Low Lift Mean?

High confidence means that the consequent appears frequently when the antecedent is present. However, a low lift means that the association may not be particularly strong compared with the normal frequency of the consequent.

This can happen when the consequent is already a popular product in the dataset. Therefore, it is useful to consider both confidence and lift when evaluating a rule.

8.3 (c) What Does a Lift Greater Than 1 Mean?

A lift greater than 1 indicates a positive association between the antecedent and consequent.

For example, the rule used in Question 6 has a lift of 3.0296. Since this value is greater than 1, it indicates a positive association between the products.

A lift equal to 1 indicates no additional association, while a lift below 1 indicates a negative association.

9 Conclusion

The Groceries dataset was analyzed to find common purchasing patterns among customers. The dataset contained 9,835 transactions and 169 different grocery items.

The analysis showed that whole milk was the most frequently purchased individual item. The two-product itemset **{other vegetables, whole milk}** had the highest support at 7.48%.

Using the Apriori algorithm, 333 frequent itemsets and 460 association rules were generated. The rules were evaluated using support, confidence, and lift.

One of the rules examined in detail was **{citrus fruit, root vegetables} → {other vegetables}**, which had a confidence of 58.62% and a lift of 3.0296.

The results show that association rule learning can be useful for understanding customer purchasing behavior. A supermarket could use these patterns for product recommendations, product placement, promotions, and inventory planning.

Finally, the rules represent associations found in the transaction data. They do not prove that purchasing one product causes another product to be purchased.